Kubernetes Interview & Viva Prep

Beginner → Advanced → Scenario-based — 80+ questions with model answers

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Beginner

Q. What is a Pod?

A. The smallest deployable unit in Kubernetes — one or more containers that share a network namespace, IPC and volumes.

Q. What is the difference between a Deployment and a ReplicaSet?

A. A ReplicaSet keeps N replicas running. A Deployment manages ReplicaSets and adds rolling updates, rollback and revision history.

Q. Why do we need a Service?

A. Pod IPs are ephemeral. A Service gives a stable virtual IP and DNS name that load-balances to current pod IPs via kube-proxy.

Q. What is kubectl?

A. The command-line client that talks to the kube-apiserver REST endpoints to read and mutate cluster state.

Q. What is the role of etcd?

A. Strongly-consistent key-value store that holds the entire cluster state. Losing etcd means losing the cluster definition.

Intermediate

Q. Explain the Pod scheduling flow.

A. Scheduler watches Pending pods, filters nodes by predicates (resources, taints, affinity), scores remaining nodes (least-loaded, image locality), binds the pod to the highest-scoring node.

Q. How does kube-proxy implement a Service?

A. It watches Service and Endpoints objects and programs iptables (or IPVS) rules on every node so that traffic to the Service ClusterIP DNATs to a healthy pod IP.

Q. What happens on kubectl apply?

A. Client merges the manifest with the last-applied annotation, sends a PATCH to the API server, which validates, runs admission webhooks, writes to etcd; controllers reconcile.

Q. What is a StatefulSet for?

A. Workloads that need stable identity and persistent storage per replica — databases, Kafka, ZooKeeper. Provides ordered start/stop and predictable DNS names.

Advanced

Q. How would you debug a node showing DiskPressure?

A. Check df -h on /var/lib/kubelet and /var/log; identify large images via crictl images; crictl rmi --prune; raise eviction thresholds only as a last resort.

Q. How does mTLS work between pods in a service mesh?

A. Sidecar (Envoy) intercepts traffic, presents a SPIFFE identity issued by the mesh CA, performs TLS handshake, enforces authorisation policies — app code is unchanged.

Q. How would you achieve blue/green with native Kubernetes?

A. Two Deployments (blue + green) with different labels; one Service selects the active colour; flip the selector to cut over; keep the old version for instant rollback.

Scenario-based / Viva

Q. A pod is stuck Pending — walk me through the investigation.

A. kubectl describe pod → look at Events. Possibilities: (a) FailedScheduling → not enough CPU/mem, no matching nodeSelector, untolerated taint, PVC not bound; (b) image pull errors → registry/secret. Verify with kubectl get nodes, kubectl get pvc, kubectl describe sa.

Q. Customers see 502 errors but pods are Running. What's your plan?

A. Layer-by-layer: kubectl get endpoints <svc> — empty? selector mismatch. Populated? kubectl exec a debug pod and curl the Service. If that works, the Ingress controller is the culprit — check its logs and the backend health checks.

Q. Your Windows pods scheduled onto Linux nodes and crashed. Why?

A. Missing nodeSelector: kubernetes.io/os: windows and tolerations for the Windows taint. Add them to the PodSpec; consider an admission webhook that injects them automatically.

Q. How would you rotate a leaked Service Account token?

A. Delete the Secret backing the token (or the bound ServiceAccount), force pods to restart so they mount the new token, audit kube-apiserver logs for any usage of the old token after rotation, rotate downstream credentials reachable with that SA.